

By modus ponens,  $r$  is True.  $\therefore xy$  is an irrational no.  $\square$

13) Prove that if  $x$  is irrational, then  $1/x$  is irrational.

Soln:  $p: x$  is irrational  $q: 1/x$  is irrational  
 $\neg q: 1/x$  is not irrational, i.e.  $1/x$  is rational (Defn. 2)  
 $\neg p: x$  is not irrational, i.e.  $x$  is rational (Defn. 2)

We shall prove  $p \rightarrow \neg q$  is True by contraposition.

$\therefore$  We shall prove  $\neg q \rightarrow \neg p$  is True.

Let,  $\neg q$  be True.  $\therefore 1/x$  is an arbitrary rational no.  
By defn. 2, there exists integers  $a, b$  where  $b \neq 0$  s.t.  $\frac{1}{x} = \frac{a}{b}$   
 $\therefore x = \frac{b}{a}$ .  $\therefore \frac{1}{x}$  exists,  $\therefore x \neq 0$ .  $\therefore b \neq 0$ . Also,  $\frac{1}{x} \neq 0$ ,  $\therefore$  no

number exists which divides 1 to get 0.  $\therefore a \neq 0$ . By defn 2,  $x$  is a rational number.  $\therefore \neg p$  is True. By defn. of conditional,  $\neg q \rightarrow \neg p$  is True.

$\therefore p \rightarrow \neg q$  is True. By modus ponens since  $p$  is True,  $q$  is True.

$\therefore 1/x$  is irrational  $\square$

14) Prove that if  $x$  is rational, and  $x \neq 0$  then  $1/x$  is rational.

Soln:  $p: x$  is rational  $q: x \neq 0$   $r: 1/x$  is rational

We shall prove  $(p \wedge q) \rightarrow r$  is True by a direct proof.

Let,  $(p \wedge q)$  be True. By defn. of AND,  $p$  is True and  $q$  is True.  
 $\therefore x$  is a rational number and  $x \neq 0$ . By defn 2, there exists integers

$a, b$  where  $b \neq 0$  such that  $x = \frac{a}{b}$ .  $\therefore x \neq 0$ ,  $\therefore a \neq 0$ .

$\therefore \frac{1}{x} = \frac{b}{a}$ . By defn 2,  $\frac{1}{x}$  is rational.  $\therefore r$  is True

By defn. of conditional,  $(p \wedge q) \rightarrow r$  is True.  $\square$